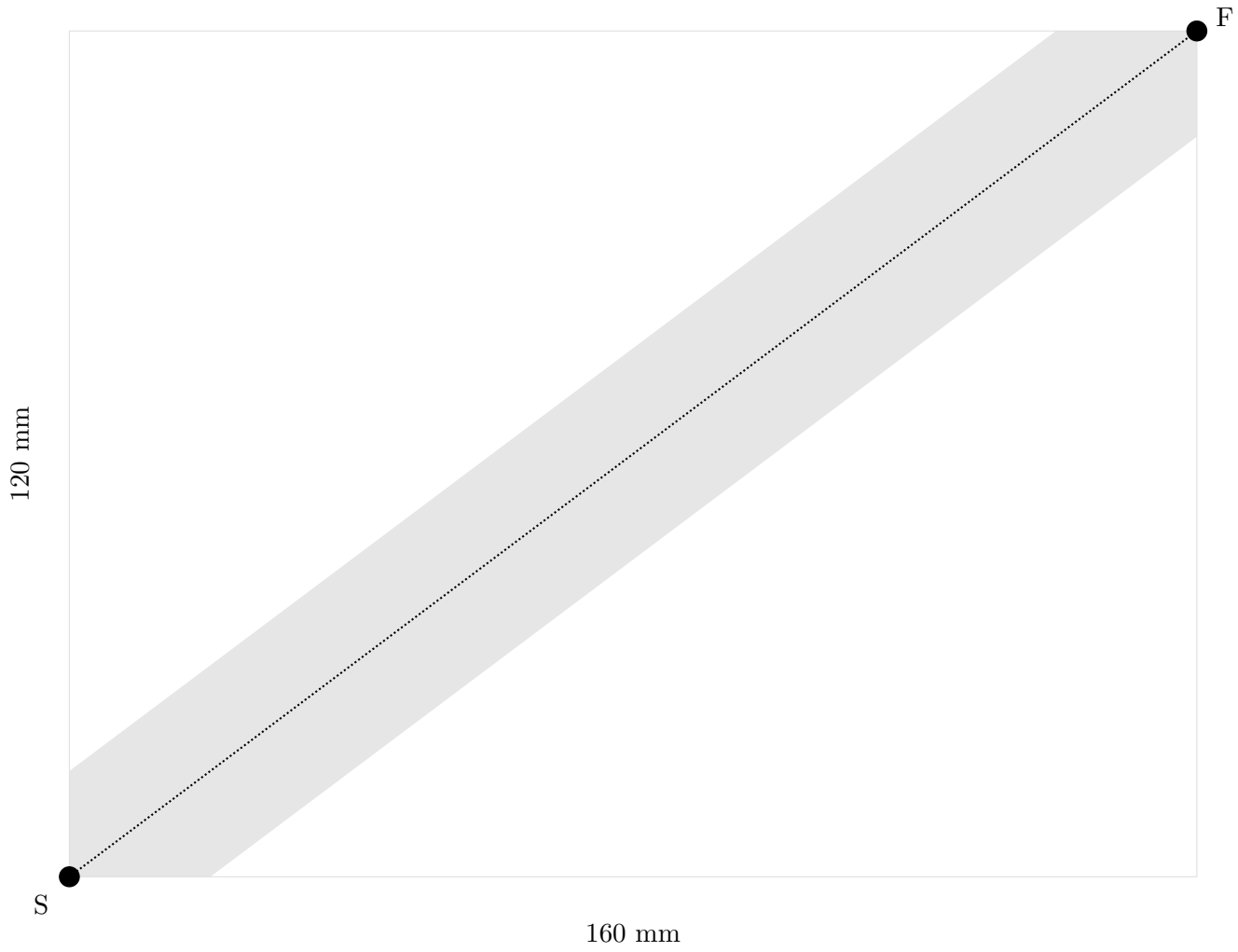
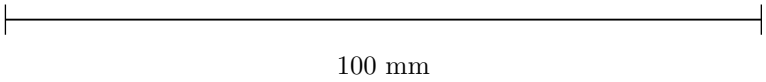
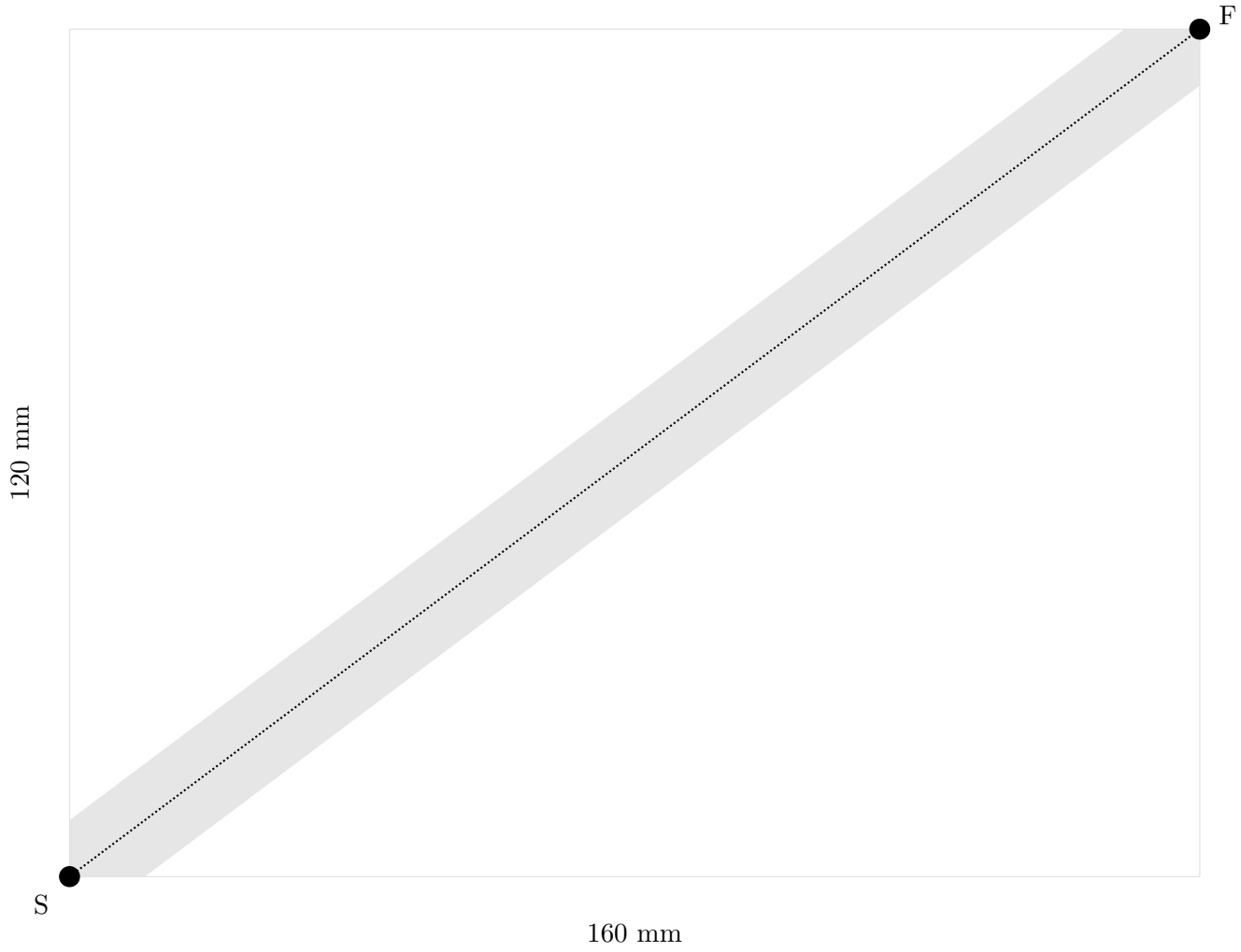
Problem 4: Find the fewest turns possible for an S-to-F staircase inside the gray strip. Its pieces may follow the strip's boundary. Explain why fewer turns cannot work.



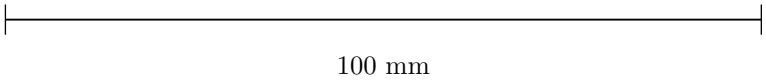
The gray strip reaches 15 mm vertically above and below the dotted diagonal, clipped by the rectangle.



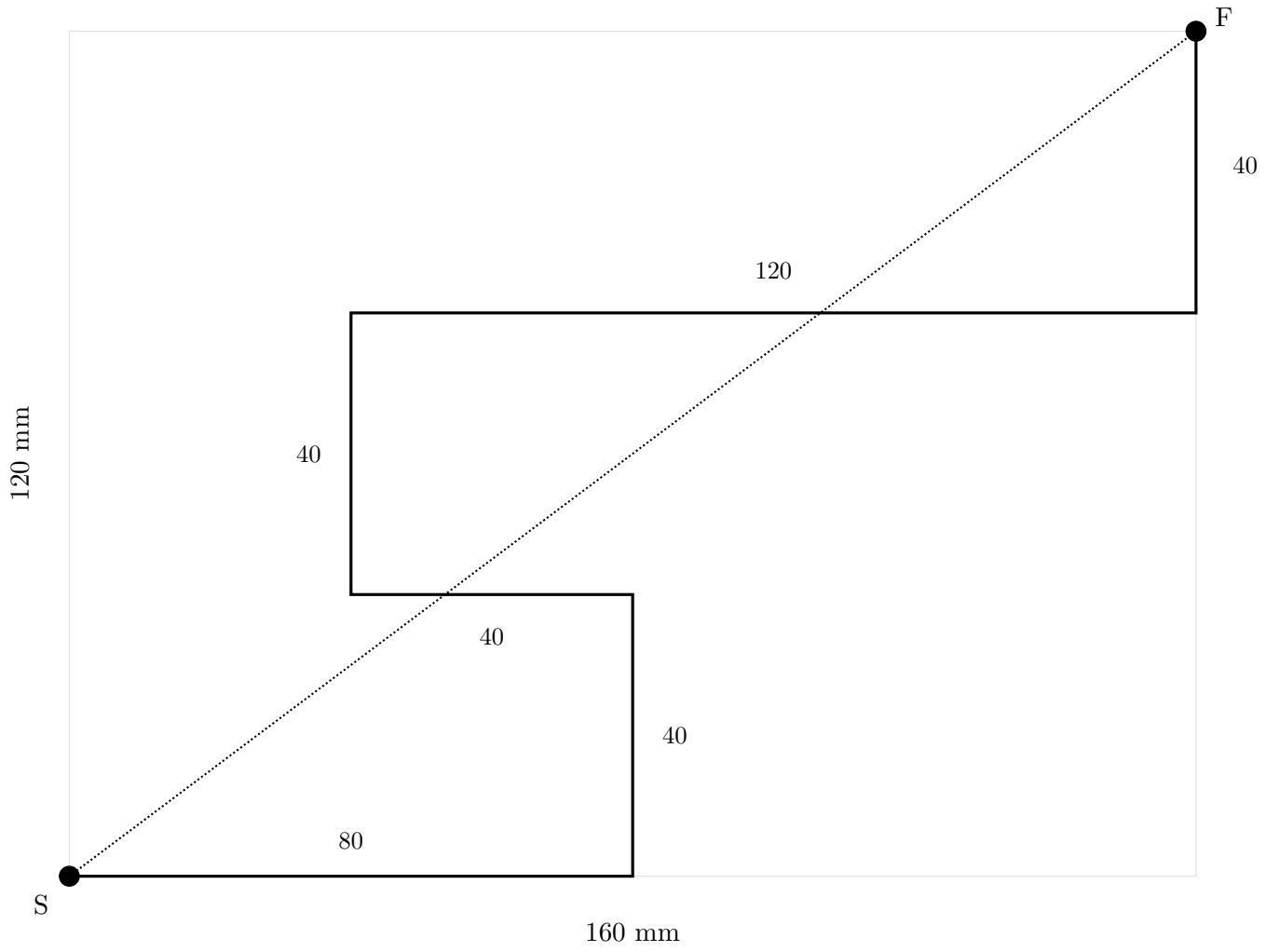
Problem 5: Fit a staircase inside this narrower gray strip. Can the new staircase be shorter than the one you drew in the wider strip? Explain your answer.



The strip reaches 8 mm vertically above and below the diagonal, clipped by the rectangle.



Problem 6: Find this path's exact length. It goes left once. Explain which part of your argument about right-and-up staircases no longer applies.



Problem 7: A staircase may use more and smaller right-and-up pieces. Could any finite number of pieces give the same length as the 200 mm diagonal? Explain which distances can shrink and which length stays the same.